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Roll. No:

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

THIRD SEMESTER-B.TECH.(ECE & EIOT)

END-SEMESTER THEORY EXAMINATION, NOV-DEC, 2024

Course Code: ECECC06/ EIECC06/ECECC303/EIECC303

Course Title: Probability Theory and Random Process

Time: 3:00 Hours

Max. Marks: 50

Note: Attempt all five questions in given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q No.	Question	Marks	CO
Q1	Attempt any two parts of the following		
1i	A lot of 100 semiconductor chips contains 20 that are defective. Two chips are selected at random, without replacement, from the lot. 1. What is the probability that the first one selected is defective? 2. What is the probability that the second one selected is defective given that the first one was defective? 3. What is the probability that both are defective?	5	CO1
1b	Calculate mean and variance of Gaussian distribution	5	CO1
1c	A noisy transmission channel has a per-digit error probability ' p ' = 0.01. 1. Calculate the probability of more than one error in 10 received digits. 2. Repeat (1), using the Poisson approximation.	5	CO1
Q2	Attempt any two parts of the following		
2a	Explain N-variate Normal random variable along with its distribution, mean and covariance matrix.	5	CO2
2b	Let X and Y be Joint continuous random variables with joint PDF is given below $f_{XY}(x,y) = \begin{cases} 2 & 0 \leq y < x < 1 \\ 0 & \text{otherwise} \end{cases}$ Their marginal density functions are given as $f_X(x) = 2x \quad 0 < x < 1 \quad \text{and}$ $f_Y(y) = 2(1-y) \quad 0 < y < 1.$ 1. Find the conditional probability density functions $f_{Y X}(y x)$ and $f_{X Y}(x y)$. 2. Find their conditional means of X and Y; $E[X Y]$ and $E[Y X]$ 3. Find covariance and correlation coefficient of X and Y.	5	CO2
2c	Let (X, Y, Z) be a trivariate random variable with joint pdf $f_{XYZ}(x,y,z) = \begin{cases} ke^{-(ax+by+cz)} & x > 0, y > 0, z > 0 \\ 0 & \text{otherwise} \end{cases}$ where a,b,c > 0 and k are constants. 1. Determine the value of k. 2. Find the marginal joint pdf of X and Y. 3. Find the marginal pdf of X. 4. Are X, Y, and Z independent?	5	CO2

Q3	Attempt any two parts of the following		
3a	Let X and Y are random variables such that $Y = aX + b$, where 'a' and 'b' are constants and $a \neq 0$. Find the CDF and PDF of Y in terms of $F_X(x)$ and $f_X(x)$. Also, if X is Gaussian distributed with zero mean and variance σ^2 , then find the values of 'a' and 'b' so that Y is Gaussian distributed with zero mean and variance = 1.	5	CO3
3b	State Central Limit theorem and explain its significance with an example.	5	CO3
3c	Let X and Y be two independent discrete random variables with the CDFs $F_X(x)$ and $F_Y(y)$. If $Z = \max(X, Y)$ and $W = \min(X, Y)$. Find the CDFs and PDFs of Z and W.	5	CO3
Q4	Attempt any two parts of the following		
4a	Let X(t) be a continuous-time WSS process with mean $\mu_X = 0$ and $R_X(\tau) = \delta(\tau)$, where $\delta(\tau)$ is the Dirac delta function. Let Y(t) be a random process defined as: $Y(t) = \int_{t-2}^t X(u) du$. Find $E[Y(t)]$ and $R_{XY}(t_1, t_2)$.	5	CO4
4b	Explain Stationarity property of Random process and thereby characterising Wide Sense Stationary (WSS) and Strict Sense Stationary (SSS) Processes.	5	CO4
4c	If $X(t) = a \cos(2\pi f_0 t + \theta)$, where 'a' and ' f_0 ' are constants and ' θ ' is uniformly distributed in the interval $(-\pi, \pi)$. Is X(t) a WSS process?	5	CO4
Q5	Attempt any two parts of the following		
5a	Explain Power Spectral Density along with its properties.	5	CO5
5b	The power spectral density of a WSS white noise process whose frequency components are limited to the range $-W < f < W$. 1. Find the average power in this process and its auto correlation function. 2. How will this results change if the above WSS process is white noise?	5	CO5
5c	A WSS random process X(t) with auto correlation $R_X(\tau) = \exp(-2a \tau)$, where 'a' is a real positive constant, Find its power spectral density. If X(t) is applied as the input to an LTI system with impulse response $h(t) = be^{-bt} u(t)$, where 'b' is real positive constant, then find the auto correlation and the power spectral density of the output Y(t).	5	CO5